



Neutrinos via Charm Decays in Astrophysical Sources

TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

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Particle Physics

- Point A
- Point B
 - Part 1
 - Part 2
- Point C
- Point D



Multimessenger Astronomy

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Astrophysical Sources

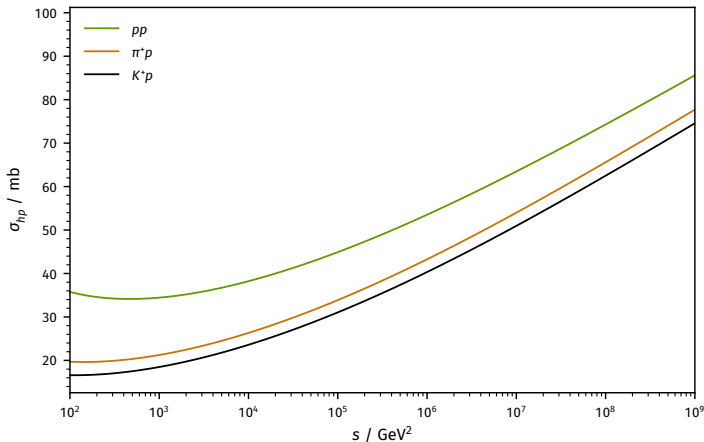
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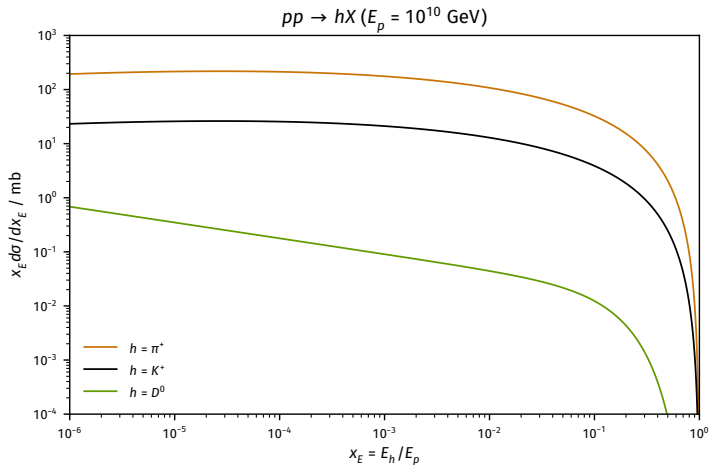
Cross Sections

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Hadron Inelastic Scattering Cross Section



Weighted Energy Differential Cross Section





Spectral Distributions

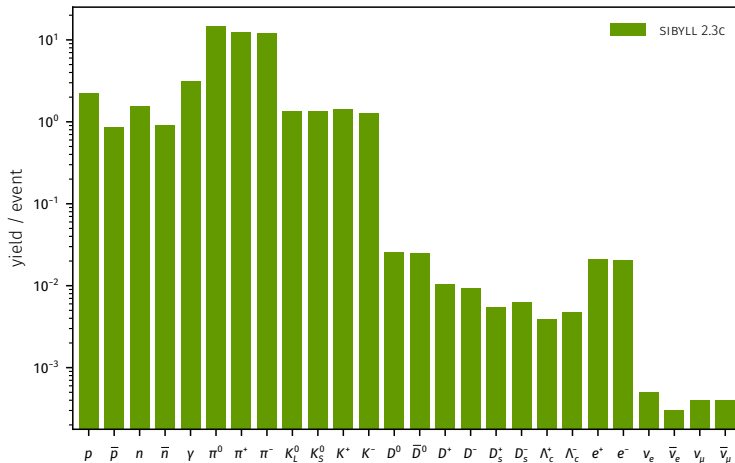
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Computation

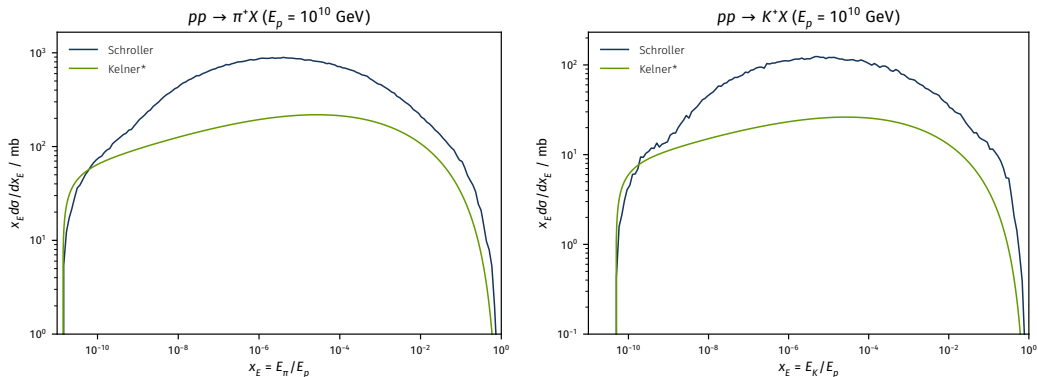
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Event Generation



Data Courtesy
M. Schroller

Weighted Energy Pion & Kaon Differential Cross Section



Derived Parametrization S. R. Kelner et. al. *Phys. Rev. D* (2006)
Event Data (SIBYLL 2.3c) Courtesy M. Schrollier



Results

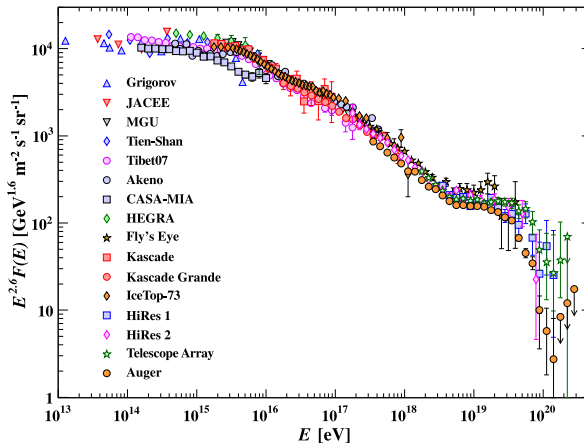
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Conclusion & Outlook

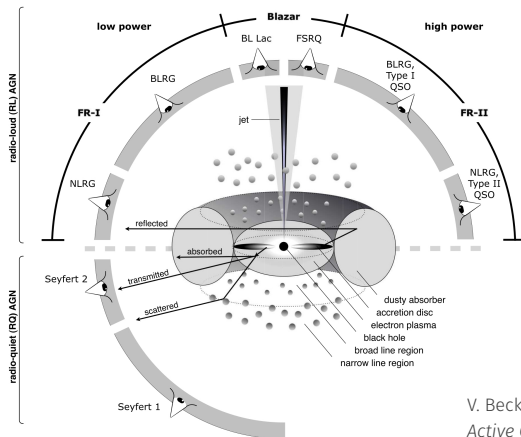
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Cosmic Ray Spectrum & Cutoff



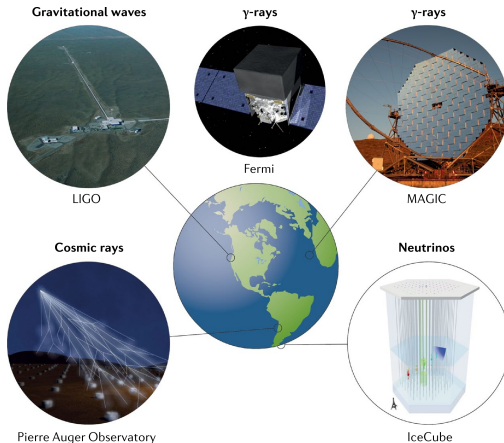
K. A. Olive et. al.
Chin. Phys. C (2014)

Unified AGN Classification Model



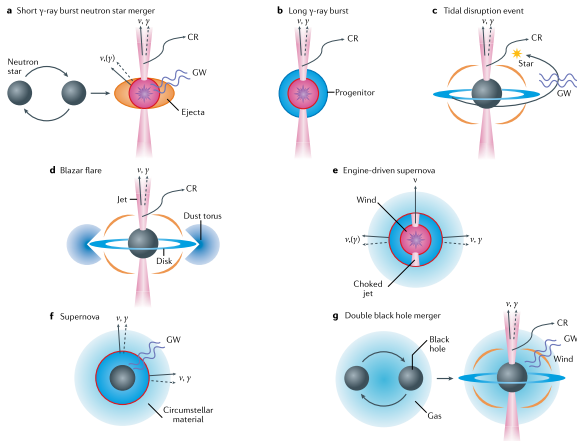
V. Beckmann, C. Shrader
Active Galactic Nuclei (2012)

Detectors



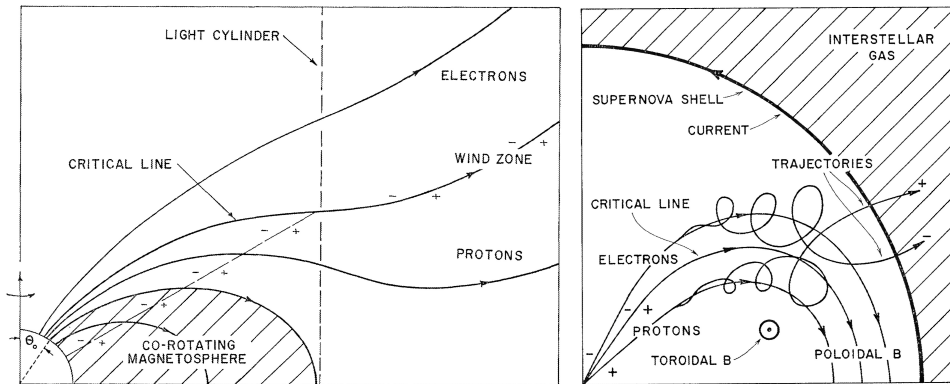
P. Mészáros et. al.
Nat. Rev. Phys. (2019)

Sources



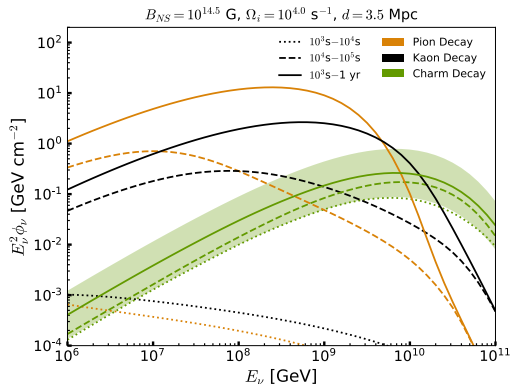
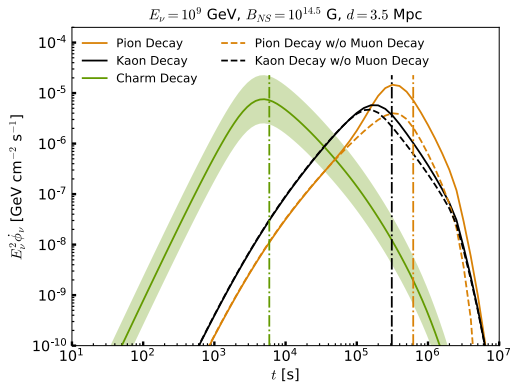
P. Mészáros et. al.
Nat. Rev. Phys. (2019)

Pulsar Particle Acceleration



P. Goldreich, W. H. Julian *ApJ* (1969)

Magnetar Neutrino Flux & Fluence Comparison



J. A. Carpio et. al. Phys. Rev. D (2020)